Scale Exercises - Section IX, Level 72

D Lydian dominant, G# altered, tritone substitution, and improvisation synthesis

A MELODIC MINOR | 11 MODULES | 640 TONES

Level 71 established the seven melodic-minor modes and their chord functions. Level 72 focuses on the most applied dominant pair inside A melodic minor: D Lydian dominant and G# altered. Their shared tritone explains direct dominant and tritone-substitute resolution into C# minor.

1. Hear D7#11

D Lydian dominant combines a complete D7 shell with the raised fourth G# and the color tones E and B.

2. Control G#7alt

G# altered contains every altered ninth and fifth around the dominant shell. The target chord explains the spelling.

3. Compare two dominants

G#7alt and D7#11 share C-F#. One is the direct dominant; the other approaches C# minor by tritone substitution.

Dominant-pair summary

```
A melodic minor      A  B  C  D  E  F# G#

D LYDIAN DOMINANT    D  E  F# G# A  B  C
relative formula     1  2  3  #4 5  6  b7
chord function       D7#11  = tritone substitute for G#7

G# ALTERED           G# A  B  C  D  E  F#
relative formula     1  b9 #9 3  b5 #5 b7
chord function       G#7alt = direct dominant of C# minor

SHARED TRITONE       C(B#) <-> F#
RESOLUTION           G#7alt -> C#m    or    D7#11 -> C#m
```

Written workload

Module	Tones
A	80
B	48
C	80
D	48
E	40
F	80
G	80
H	64
I	32
J	64
K	24
TOTAL	640

Session protocol

Choose one mode route, one dominant-resolution cell, and one progression application. State the dominant chord and destination before playing. Add one tension at a time; remove notes whenever the direct or tritone-substitute function becomes unclear.

SPELLING RULE C is B# when it functions as the major third of G#7. It remains C when it is the flat seventh of D7.

Scale Exercises - Section IX, Level 72

Dominant-pair atlas, shared tritone, and destination voice leading

DIRECT G#7alt | SUBSTITUTE D7#11 | DESTINATION C#m

THE SAME C-F# TRITONE CHANGES NAME AND RESOLUTION DIRECTION WITH THE ROOT

D7#11 - D Lydian dominant

```
Chord tones:  D   F#  A   C
Roles:        R   3   5   b7
Mode colors:  E   G#  B
Roles:        9   #11 13

Primary motion into C#m:
D -> C#   F# -> E   C -> B   G# remains common
```

G#7alt - G# altered

```
Chord shell:  G#  B#  F#
Roles:        R   3   b7
Alterations:  A   B   D   E
Roles:        b9  #9  b5  #5

Primary motion into C#m:
B# -> C#   F# -> E   A -> G#   E remains common
```

Enharmonic tritone and destination map

Pitch pair	Over G#7alt	Over D7#11	Resolution into C#m
C / B# and F#	3 and b7	b7 and 3	B#->C# / C->B; F#->E
G#	root	#11	common tone: fifth of C#m
A	b9	5	A->G#
B	#9	13	common tone: b7 of C#m
E	#5	9	common tone: b3 of C#m

Listening order

1. Play C-F# alone. 2. Add G# below it and hear direct dominant pull. 3. Add D below it and hear the tritone substitute. 4. Resolve both versions to C#m. The shell is shared; the root and voice-leading direction define the function.

Scale Exercises - Section IX, Level 72

Module A - D Lydian dominant mirrors through five positions

48 TONES | P1-P3

D-E-F#-G#-A-B-C-D | 1-2-3-#4-5-6-b7-1

Each cell ascends one octave, re-attacks the upper D, and reverses exactly. First accent D-F#-A-C. Next accent G# alone. Finally remove accents while retaining both D7 and #11 color.

P1

D7 chord tones + characteristic #4.

```
Nte|D---E---F#--G#--A---B---C---D---|D---C---B---A---G#--F#--E---D---
Deg|1---2---3---#4--5---6---b7--1---|1---b7--6---5---#4--3---2---1---
  G|-----------------------5---7---|7---5---------------------------
  D|------------6---7---9-----------|--------9---7---6--------------
  A|5---7---9----------------------|--------------------9---7---5---
```

P2

D7 chord tones + characteristic #4.

```
Nte|D---E---F#--G#--A---B---C---D---|D---C---B---A---G#--F#--E---D---
Deg|1---2---3---#4--5---6---b7--1---|1---b7--6---5---#4--3---2---1---
  G|---------------------------7---|7---------------------------------
  D|----------------7---9---10------|----10--9---7-------------------
  A|----7---9---11------------------|---------------11--9---7-------
  E|10-----------------------------|-------------------------10--
```

P3

D7 chord tones + characteristic #4.

```
Nte|D---E---F#--G#--A---B---C---D---|D---C---B---A---G#--F#--E---D---
Deg|1---2---3---#4--5---6---b7--1---|1---b7--6---5---#4--3---2---1---
  D|--------------------9---10--12--|12--10--9-----------------------
  A|--------9---11--12--------------|------------12--11--9-----------
  E|10--12-------------------------|----------------------12--10--
```

Scale Exercises - Section IX, Level 72

Module A - D Lydian dominant mirrors through five positions

32 TONES | P4-P5

D-E-F#-G#-A-B-C-D | 1-2-3-#4-5-6-b7-1

Each cell ascends one octave, re-attacks the upper D, and reverses exactly. First accent D-F#-A-C. Next accent G# alone. Finally remove accents while retaining both D7 and #11 color.

P4

D7 chord tones + characteristic #4.

```
Nte|D---E---F#--G#--A---B---C---D---|D---C---B---A---G#--F#--E---D---
Deg|1---2---3---#4--5---6---b7--1---|1---b7--6---5---#4--3---2---1---
  D|-----------------------10--12--|12--10-------------------------
  A|------------11--12--14----------|--------14--12--11--------------
  E|10--12--14----------------------|--------------------14--12--10--
```

P5

D7 chord tones + characteristic #4.

```
Nte|D---E---F#--G#--A---B---C---D---|D---C---B---A---G#--F#--E---D---
Deg|1---2---3---#4--5---6---b7--1---|1---b7--6---5---#4--3---2---1---
  B|-----------------------13--15--|15--13-------------------------
  G|------------13--14--16----------|--------16--14--13--------------
  D|12--14--16----------------------|--------------------16--14--12--
```

Position translation assignment

Take the P1 rhythm and articulation through P2-P5 without changing the degree order. Then begin every cell from the third F# rather than the root D and preserve the D7#11 function.

Scale Exercises - Section IX, Level 72

Module B - D7#11 color and resolution cells

6 CELLS | 48 TONES

NAME THE COLOR NOTE, THEN NAME ITS CHORD-TONE DESTINATION

The printed cells isolate #11, 9, 13, and b7 behavior over D7. The final note of each phrase is a D7 chord tone. Play the cell over a D7 drone before placing it inside a progression.

B1 - #11 to 5

Eight picked tones; target the final role.

```
Nte|G#---A----F#---D----E----F#---G#---A----
Deg|#4---5----3----1----2----3----#4---5----
  D|6----7-----------------------6----7----
  A|----------9----5----7----9-------------
```

B2 - #11 to 3

Eight picked tones; target the final role.

```
Nte|G#---F#---E----D----E----G#---A----F#---
Deg|#4---3----2----1----2----#4---5----3----
  D|-------------------------------7---------
  A|11---9----7---------7----11--------9----
  E|--------------10-----------------------
```

B3 - 9 to root

Eight picked tones; target the final role.

```
Nte|E----D----F#---A----C----B----A----D----
Deg|2----1----3----5----b7---6----5----1----
  D|-------------------10---9--------------
  A|----------9----12------------12--------
  E|12---10----------------------------10---
```

B4 - 13 to 5

Eight picked tones; target the final role.

```
Nte|B----A----G#---F#---E----C----B----A----
Deg|6----5----#4---3----2----b7---6----5----
  D|-----------------------10-------------
  A|14---12---11-----------------14---12---
  E|--------------14---12-----------------
```

B5 - b7 to 3

Eight picked tones; target the final role.

```
Nte|C----B----A----G#---C----A----G#---F#---
Deg|b7---6----5----#4---b7---5----#4---3----
  B|13-----------------13-----------------
  G|-----16---14---13--------14---13--------
  D|-----------------------------------16---
```

B6 - complete color line

Eight picked tones; target the final role.

```
Nte|D----E----F#---G#---A----B----C----D----
Deg|1----2----3----#4---5----6----b7---1----
  G|-----------------------------5---------
  D|--------------6----7----9--------------
  A|5----7----9-----------------------5----
```

Tension budget

Practice #11, 9, and 13 separately. Combine two colors only after each can be heard and resolved on command. If the phrase sounds like a scale run, reduce it to color-target-color-target.

Scale Exercises - Section IX, Level 72

Module C - G# altered mirrors through five registers

48 TONES | LOWER ROUTES

G#-A-B-C-D-E-F#-G# | 1-b9-#9-3-b5-#5-b7-1

C is heard as B#, the major third of G#7. Play the root, major third, and flat seventh first. Add one alteration at a time, then restore the complete eight-note route.

R1 lower

Direct dominant of C# minor.

```
Nte|G#--A---B---B#--D---E---F#--G#--|G#--F#--E---D---B#--B---A---G#--
Deg|1---b9--#9--3---b5--#5--b7--1---|1---b7--#5--b5--3---#9--b9--1---
  A|--------------------7---9---11--|11--9---7----------------------
  E|4---5---7---8---10--------------|------------10--8---7---5---4---
```

R2 low-middle

Direct dominant of C# minor.

```
Nte|G#--A---B---B#--D---E---F#--G#--|G#--F#--E---D---B#--B---A---G#--
Deg|1---b9--#9--3---b5--#5--b7--1---|1---b7--#5--b5--3---#9--b9--1---
  G|--------------------9---11--13--|13--11--9----------------------
  D|6---7---9---10--12--------------|------------12--10--9---7---6---
```

R3 middle

Direct dominant of C# minor.

```
Nte|G#--A---B---B#--D---E---F#--G#--|G#--F#--E---D---B#--B---A---G#--
Deg|1---b9--#9--3---b5--#5--b7--1---|1---b7--#5--b5--3---#9--b9--1---
  G|----------------------------13--|13----------------------------
  D|--------------------14--16------|----16--14----------------------
  A|11--12--14--15--17--------------|------------17--15--14--12--11--
```

Scale Exercises - Section IX, Level 72

Module C - G# altered mirrors through five registers

32 TONES | UPPER ROUTES

G#-A-B-C-D-E-F#-G# | 1-b9-#9-3-b5-#5-b7-1

C is heard as B#, the major third of G#7. Play the root, major third, and flat seventh first. Add one alteration at a time, then restore the complete eight-note route.

R4 upper-middle

Direct dominant of C# minor.

```
Nte|G#--A---B---B#--D---E---F#--G#--|G#--F#--E---D---B#--B---A---G#--
Deg|1---b9--#9--3---b5--#5--b7--1---|1---b7--#5--b5--3---#9--b9--1---
  e|-----------------------14--16--|16--14-------------------------
  B|---------------15--17----------|--------17--15------------------
  G|13--14--16--17------------------|----------------17--16--14--13--
```

R5 high

Direct dominant of C# minor.

```
Nte|G#--A---B---B#--D---E---F#--G#--|G#--F#--E---D---B#--B---A---G#--
Deg|1---b9--#9--3---b5--#5--b7--1---|1---b7--#5--b5--3---#9--b9--1---
  e|-------------------12--14--16--|16--14--12----------------------
  B|9---10--12--13--15--------------|------------15--13--12--10--9---
```

Altered-scale stop rule

If G#7 loses its major third, stop and play G#-B#-F# alone. The alterations are meaningful only around a clearly audible dominant shell and destination.

Scale Exercises - Section IX, Level 72

Module D - G#7alt tension resolution into C# minor

6 CELLS | 48 TONES

FOUR ALTERED-DOMINANT TONES -> FOUR C#m7 TONES

Every cell crosses the dominant-to-tonic barline. The first four tones expose one altered color; the second four establish C#m7. Preserve the written resolution direction.

D1 - b9 and major 3 to tonic

G#7alt | C#m7

```
Nte|G#--A---G#--B#--|C#--E---G#--B---
Rol|R---b9--R---3---|R---b3--5---b7--
  D|----------------|--------6---9---
  A|----------------|4---7-----------
  E|4---5---4---8---|----------------
```

D2 - #9 and b7 to tonic

G#7alt | C#m7

```
Nte|G#--B---B#--F#--|C#--E---G#--B---
Rol|R---#9--3---b7--|R---b3--5---b7--
  G|------------11--|------------4---
  D|6---9---10------|--------6-------
  A|----------------|----7-----------
  E|----------------|9---------------
```

D3 - b5 to tonic root

G#7alt | C#m7

```
Nte|G#--D---B#--F#--|C#--E---G#--B---
Rol|R---b5--3---b7--|R---b3--5---b7--
  B|----------------|------------12--
  G|----------------|----9---13------
  D|------------16--|11--------------
  A|11--17--15------|----------------
```

D4 - #5 as common minor-third color

G#7alt | C#m7

```
Nte|G#--E---D---B#--|C#--E---G#--B---
Rol|R---#5--b5--3---|R---b3--5---b7--
  B|----17--15------|----------------
  G|13----------17--|--------13--16--
  D|----------------|----14----------
  A|----------------|16--------------
```

D5 - b7 falling to tonic b3

G#7alt | C#m7

```
Nte|G#--F#--E---D---|C#--E---G#--B---
Rol|R---b7--#5--b5--|R---b3--5---b7--
  e|----14--12------|----12--16------
  B|9-----------15--|14--------------
  G|----------------|------------16--
```

D6 - major 3 rising to tonic root

G#7alt | C#m7

```
Nte|A---B---B#--G#--|C#--E---G#--B---
Rol|b9--#9--3---R---|R---b3--5---b7--
  D|----------------|--------6---9---
  A|----------------|4---7-----------
  E|5---7---8---4---|----------------
```

Destination audit

Remove the first four notes and play C#m7 alone. Then restore the dominant. The altered phrase succeeds only if the tonic sounds more stable and the barline remains clear.

Scale Exercises - Section IX, Level 72

Module E - shared tritone shells and enharmonic spelling

5 REGISTERS | 40 TONES

G#7alt: B#-F# | D7#11: C-F# | SAME PITCHES, DIFFERENT FUNCTION

E1 - low register

G#7alt shell | D7 shell

```
Nte|G#---B#---F#---B#---|D----F#---C----F#---
Rol|R----3----b7---3----|R----3----b7---3----
  A|----------9---------|5----9--------9----
  E|4----8--------8----|----------8--------
```

E2 - low-middle

G#7alt shell | D7 shell

```
Nte|G#---B#---F#---B#---|D----F#---C----F#---
Rol|R----3----b7---3----|R----3----b7---3----
  G|----------11--------|7----11--------11---
  D|6----10--------10---|----------10-------
```

E3 - middle

G#7alt shell | D7 shell

```
Nte|G#---B#---F#---B#---|D----F#---C----F#---
Rol|R----3----b7---3----|R----3----b7---3----
  D|----------16--------|12---16--------16---
  A|11---15--------15---|----------15-------
```

E4 - upper-middle

G#7alt shell | D7 shell

```
Nte|G#---B#---F#---B#---|D----F#---C----F#---
Rol|R----3----b7---3----|R----3----b7---3----
  e|----------14--------|-----14--------14---
  B|--------------------|15-----------------
  G|13---17--------17---|----------17-------
```

E5 - high

G#7alt shell | D7 shell

```
Nte|G#---B#---F#---B#---|D----F#---C----F#---
Rol|R----3----b7---3----|R----3----b7---3----
  e|----------14--------|-----14--------14---
  B|9----13--------13---|15--------13-------
```

Root test

Have a partner or loop alternate G# and D beneath the same C-F# dyad. Identify the chord function by ear before resolving both versions to C# minor.

Scale Exercises - Section IX, Level 72

Module F - direct dominant versus tritone-substitute cadence pairs

2 PAIRS | 32 TONES

G#7alt -> C#m7 | D7#11 -> C#m7

Each pair uses the same C#m7 destination twice. The direct dominant exposes b9 and the major third. The substitute exposes the root D, #11 G#, and the shared C-F# tritone.

F1 - R1 lower

G#7alt | C#m7 | D7#11 | C#m7

```
Nte|G#--A---B#--F#--|C#--E---G#--B---|D---G#--F#--C---|C#--E---G#--B---|
Rol|R---b9--3---b7--|R---b3--5---b7--|R---#11-3---b7--|R---b3--5---b7--|
  G|----------------|----------------|------------5---|----------------|
  D|----------------|--------6---9---|----6-----------|--------6---9---|
  A|------------9---|4---7-----------|5-------9-------|4---7-----------|
  E|4---5---8-------|----------------|----------------|----------------|
```

F2 - R2 low-middle

G#7alt | C#m7 | D7#11 | C#m7

```
Nte|G#--A---B#--F#--|C#--E---G#--B---|D---G#--F#--C---|C#--E---G#--B---|
Rol|R---b9--3---b7--|R---b3--5---b7--|R---#11-3---b7--|R---b3--5---b7--|
  G|------------11--|------------4---|----------------|------------4---|
  D|6---7---10------|--------6-------|------------10--|--------6-------|
  A|----------------|----7-----------|----11--9-------|----7-----------|
  E|----------------|9---------------|10--------------|9---------------|
```

Scale Exercises - Section IX, Level 72

Module F - direct dominant versus tritone-substitute cadence pairs

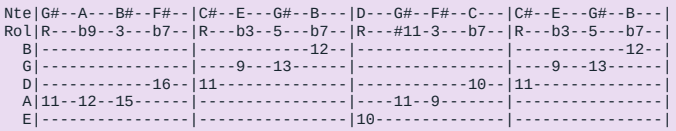
2 PAIRS | 32 TONES

G#7alt -> C#m7 | D7#11 -> C#m7

Each pair uses the same C#m7 destination twice. The direct dominant exposes b9 and the major third. The substitute exposes the root D, #11 G#, and the shared C-F# tritone.

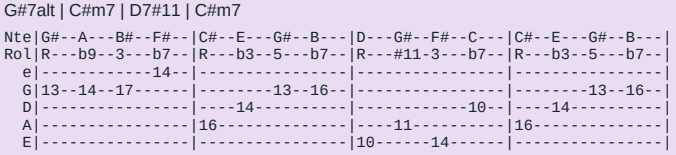
F3 - R3 middle

G#7alt | C#m7 | D7#11 | C#m7



F4 - R4 upper-middle

G#7alt | C#m7 | D7#11 | C#m7



Scale Exercises - Section IX, Level 72

Module F - direct dominant versus tritone-substitute cadence pairs

1 PAIR | 16 TONES

G#7alt -> C#m7 | D7#11 -> C#m7

Each pair uses the same C#m7 destination twice. The direct dominant exposes b9 and the major third. The substitute exposes the root D, #11 G#, and the shared C-F# tritone.

F5 - R5 high

G#7alt | C#m7 | D7#11 | C#m7

```
Nte|G#--A---B#--F#--|C#--E---G#--B---|D---G#--F#--C---|C#--E---G#--B---|
Rol|R---b9--3---b7--|R---b3--5---b7--|R---#11-3---b7--|R---b3--5---b7--|
  e|------------14--|----12--16------|----------------|----12--16------|
  B|9---10--13------|14--------------|------------13--|14--------------|
  G|----------------|------------16--|----13----------|------------16--|
  D|----------------|----------------|12------16------|----------------|
```

Cadence comparison assignment

1. Play both cadences at equal volume. 2. Sustain the shared C-F# tritone before adding the roots. 3. Remove all color tones and compare only shells. 4. Improvise one direct-dominant answer and one tritone-substitute answer with the same rhythm and destination.

Scale Exercises - Section IX, Level 72

Module G - applied-dominant phrases in C# minor

3 PHRASES | 48 TONES

| C#m7 | F#m7 | G#7alt | C#m7 | = i-iv-V-i

The first note of each bar is a chord tone. C is spelled B# in the G#7alt measure. After the quarter-note pass, replace notes 2-4 while keeping the beat-one targets and dominant resolution fixed.

R1

Four bars; 16 chord-oriented tones.

```
Nte|C#--E---G#--B---|F#--A---C#--E---|G#--B#--F#--A---|B---G#--E---C#--|
Rol|>R--b3--5---b7--|>R--b3--5---b7--|>R--3---b7--b9--|>b7-5---b3--R---|
  D|--------6---9---|----------------|----------------|9---6-----------|
  A|4---7-----------|--------4---7---|--------9-------|--------7---4---|
  E|----------------|2---5-----------|4---8-------5---|----------------|
```

R2

Four bars; 16 chord-oriented tones.

```
Nte|C#--E---G#--B---|F#--A---C#--E---|G#--B#--F#--A---|B---G#--E---C#--|
Rol|>R--b3--5---b7--|>R--b3--5---b7--|>R--3---b7--b9--|>b7-5---b3--R---|
  G|------------4---|------------9---|--------11------|4---------------|
  D|--------6-------|----7---11------|6---10------7---|----6-----------|
  A|----7-----------|9---------------|----------------|--------7-------|
  E|9---------------|----------------|----------------|------------9---|
```

R3

Four bars; 16 chord-oriented tones.

```
Nte|C#--E---G#--B---|F#--A---C#--E---|G#--B#--F#--A---|B---G#--E---C#--|
Rol|>R--b3--5---b7--|>R--b3--5---b7--|>R--3---b7--b9--|>b7-5---b3--R---|
  B|------------12--|----------------|----------------|12--------------|
  G|----9---13------|----------------|----------------|----13--9-------|
  D|11--------------|------------14--|--------16------|------------11--|
  A|----------------|----12--16------|11--15------12--|----------------|
  E|----------------|14--------------|----------------|----------------|
```

Scale Exercises - Section IX, Level 72

Module G - applied-dominant phrases in C# minor

2 PHRASES | 32 TONES

| C#m7 | F#m7 | G#7alt | C#m7 | = i-iv-V-i

The first note of each bar is a chord tone. C is spelled B# in the G#7alt measure. After the quarter-note pass, replace notes 2-4 while keeping the beat-one targets and dominant resolution fixed.

R4

Four bars; 16 chord-oriented tones.

```
Nte|C#--E---G#--B---|F#--A---C#--E---|G#--B#--F#--A---|B---G#--E---C#--|
Rol|>R--b3--5---b7--|>R--b3--5---b7--|>R--3---b7--b9--|>b7-5---b3--R---|
  e|----------------|------------12--|--------14------|----------------|
  B|----------------|--------14------|----------------|----------------|
  G|--------13--16--|----14----------|13--17------14--|16--13----------|
  D|----14----------|16--------------|----------------|--------14------|
  A|16--------------|----------------|----------------|------------16--|
```

R5

Four bars; 16 chord-oriented tones.

```
Nte|C#--E---G#--B---|F#--A---C#--E---|G#--B#--F#--A---|B---G#--E---C#--|
Rol|>R--b3--5---b7--|>R--b3--5---b7--|>R--3---b7--b9--|>b7-5---b3--R---|
  e|----12--16------|------------12--|--------14------|----16--12------|
  B|14--------------|7---10--14------|9---13------10--|------------14--|
  G|------------16--|----------------|----------------|16--------------|
```

Scale Exercises - Section IX, Level 72

Module H - parent, applied, substitute, and minor-cadence comparisons

4 PHRASES | 64 TONES

FOUR HARMONIC JOBS; ONE FOUR-BAR PHRASE FOR EACH

The parent E7b13 returns to A minor. G#7alt and D7#11 both resolve to C# minor. The final phrase adds the actual ii-m7b5-V-i cadence in C# minor.

H1 - parent dominant to A minor

Four bars; four tones per bar.

```
Nte|B---D---F#--A---|E---G#--D---C---|A---C---E---G#--|A---C---E---F#--|
Rol|>R--b3--5---b7--|>R--3---b7--b13-|>R--b3--5---7---|>R--b3--5---6---|
  G|----------------|--------7-------|----------------|----------------|
  D|------------7---|----6-------10--|------------6---|----------------|
  A|----5---9-------|7---------------|--------7-------|--------7---9---|
  E|7---------------|----------------|5---8-----------|5---8-----------|
```

H2 - direct applied dominant

Four bars; four tones per bar.

```
Nte|C#--E---G#--B---|F#--A---C#--E---|G#--B#--F#--A---|C#--E---G#--B---|
Rol|>R--b3--5---b7--|>R--b3--5---b7--|>R--3---b7--b9--|>R--b3--5---b7--|
  G|------------4---|------------9---|--------11------|------------4---|
  D|--------6-------|----7---11------|6---10------7---|--------6-------|
  A|----7-----------|9---------------|----------------|----7-----------|
  E|9---------------|----------------|----------------|9---------------|
```

H3 - tritone-substitute dominant

Four bars; four tones per bar.

```
Nte|C#--E---G#--B---|F#--A---C#--E---|D---F#--C---G#--|C#--E---G#--B---|
Rol|>R--b3--5---b7--|>R--b3--5---b7--|>R--3---b7--#11-|>R--b3--5---b7--|
  B|------------12--|----------------|----------------|------------12--|
  G|----9---13------|----------------|--------5-------|----9---13------|
  D|11--------------|------------14--|------------6---|11--------------|
  A|----------------|----12--16------|5---9-----------|----------------|
  E|----------------|14--------------|----------------|----------------|
```

H4 - minor ii-m7b5-V-i

Four bars; four tones per bar.

```
Nte|D#--F#--A---C#--|G#--B#--F#--A---|C#--E---G#--B#--|C#--E---G#--B---|
Rol|>R--b3--b5--b7--|>R--3---b7--b9--|>R--b3--5---7---|>R--b3--5---b7--|
  G|------------6---|----------------|----------------|----------------|
  D|--------7-------|----------------|--------6---10--|--------6---9---|
  A|6---9-----------|--------9-------|4---7-----------|4---7-----------|
  E|----------------|4---8-------5---|----------------|----------------|
```

Scale Exercises - Section IX, Level 72

Module I - transposed tritone-substitution guide-tone cells

4 DESTINATIONS | 32 TONES

DIRECT DOMINANT TRITONE -> SUBSTITUTE TRITONE -> TONIC GUIDE TONES

The A-melodic-minor pair provides the master model. These four cells transpose the same direct/substitute relationship to C major, F major, A minor, and D minor.

I1 - G7 / Db7 -> Cmaj7

Direct shell | substitute shell | tonic tones

```
Nte|B----F----|F----Cb---|E----B----C----G----
Rol|3----b7---|3----b7---|3----7----R----5----
  G|4---------|-----4----|-----4--------------
  D|-----3----|3---------|2--------------5----
  A|----------|----------|----------3---------
```

I2 - C7 / Gb7 -> Fmaj7

Direct shell | substitute shell | tonic tones

```
Nte|E----Bb---|Bb---Fb---|A----E----F----C----
Rol|3----b7---|3----b7---|3----7----R----5----
  e|----------|----------|5-------------------
  B|5---------|-----5----|-----5----6---------
  G|-----3----|3---------|---------------5----
```

I3 - E7 / Bb7 -> Am7

Direct shell | substitute shell | tonic tones

```
Nte|G#---D----|D----Ab---|C----G----A----E----
Rol|3----b7---|3----b7---|b3---b7---R----5----
  B|9---------|-----9----|-----8---------5----
  G|-----7----|7---------|5-------------------
  D|----------|----------|----------7---------
```

I4 - A7 / Eb7 -> Dm7

Direct shell | substitute shell | tonic tones

```
Nte|C#---G----|G----Db---|F----C----D----A----
Rol|3----b7---|3----b7---|b3---b7---R----5----
  B|-----8----|8---------|6-------------------
  G|6---------|-----6----|-----5----7---------
  D|----------|----------|---------------7----
```

Scale Exercises - Section IX, Level 72

Module J - sixteen-bar melodic-minor dominant-synthesis solo, bars 1-8

32 TONES | FIRST HALF

FOUR QUARTER-NOTE EVENTS PER BAR; BEAT 1 IS A CHORD TONE

The form moves from parent A melodic minor into C# minor, then compares direct and tritone-substitute dominants before closing with ii-m7b5-V-i. Add one articulation per bar only after the target schedule is secure.

Bar 1 - AmMaj7

Four chord-oriented tones.

```
Nte|A---C---E---G#--
Rol|>R--b3--5---7---
  D|------------6---
  A|--------7-------
  E|5---8-----------
```

Bar 2 - D7#11

Four chord-oriented tones.

```
Nte|D---F#--G#--C---
Rol|>R--3---#11-b7--
  G|------------5---
  D|--------6-------
  A|5---9-----------
```

Bar 3 - E7b13

Four chord-oriented tones.

```
Nte|E---G#--D---C---
Rol|>R--3---b7--b13-
  G|--------7-------
  D|----6-------10--
  A|7---------------
```

Bar 4 - AmMaj7

Four chord-oriented tones.

```
Nte|A---G#--E---C---
Rol|>R--7---5---b3--
  D|7---6-----------
  A|--------7-------
  E|------------8---
```

Bar 5 - C#m7

Four chord-oriented tones.

```
Nte|C#--E---G#--B---
Rol|>R--b3--5---b7--
  G|------------4---
  D|--------6-------
  A|----7-----------
  E|9---------------
```

Bar 6 - F#m7

Four chord-oriented tones.

```
Nte|F#--A---C#--E---
Rol|>R--b3--5---b7--
  G|------------9---
  D|----7---11------
  A|9---------------
```

Bar 7 - G#7alt

Four chord-oriented tones.

```
Nte|G#--A---B#--F#--
Rol|>R--b9--3---b7--
  G|------------11--
  D|6---7---10------
```

Bar 8 - C#m7

Four chord-oriented tones.

```
Nte|B---G#--E---C#--
Rol|>b7-5---b3--R---
  G|4---------------
  D|----6-----------
  A|--------7-------
  E|------------9---
```

Scale Exercises - Section IX, Level 72

Module J - sixteen-bar melodic-minor dominant-synthesis solo, bars 9-16

32 TONES | SECOND HALF

FOUR QUARTER-NOTE EVENTS PER BAR; BEAT 1 IS A CHORD TONE

The form moves from parent A melodic minor into C# minor, then compares direct and tritone-substitute dominants before closing with ii-m7b5-V-i. Add one articulation per bar only after the target schedule is secure.

Bar 9 - C#m7

Four chord-oriented tones.

```
Nte|C#--E---G#--B---
Rol|>R--b3--5---b7--
  G|--------13--16--
  D|----14----------
  A|16--------------
```

Bar 10 - F#m7

Four chord-oriented tones.

```
Nte|F#--A---C#--E---
Rol|>R--b3--5---b7--
  e|------------12--
  B|--------14------
  G|----14----------
  D|16--------------
```

Bar 11 - D7#11

Four chord-oriented tones.

```
Nte|D---F#--G#--C---
Rol|>R--3---#11-b7--
  B|------------13--
  G|--------13------
  D|12--16----------
```

Bar 12 - C#m7

Four chord-oriented tones.

```
Nte|B---G#--E---C#--
Rol|>b7-5---b3--R---
  G|16--13----------
  D|--------14------
  A|------------16--
```

Bar 13 - D#m7b5

Four chord-oriented tones.

```
Nte|D#--F#--A---C#--
Rol|>R--b3--b5--b7--
  B|----7---10--14--
  G|8---------------
```

Bar 14 - G#7alt

Four chord-oriented tones.

```
Nte|G#--B---B#--F#--
Rol|>R--#9--3---b7--
  e|------------14--
  G|13--16--17------
```

Bar 15 - C#mMaj7

Four chord-oriented tones.

```
Nte|C#--E---G#--B#--
Rol|>R--b3--5---7---
  G|--------13--17--
  D|----14----------
  A|16--------------
```

Bar 16 - C#m7

Four chord-oriented tones.

```
Nte|C#--E---G#--B---
Rol|>R--b3--5---b7--
  G|--------13--16--
  D|----14----------
  A|16--------------
```

Bar 15 uses B# as the major seventh of C#mMaj7; bar 16 lowers it to B for C#m7 closure.

Standard tuning. Parent collection: A melodic minor. Dominant functions are named by chord and destination.

Scale Exercises - Section IX, Level 72

Module K - recovery cards and three-pass improvisation capstone

12 CARDS | 24 TONES

PRESERVE THE PULSE; RESOLVE TO THE NEAREST SECURE FUNCTION

Six cards repair D7#11 color and six repair G#7alt tension. Play each pair once as written, then insert it into the appropriate dominant bar without restarting the phrase.

D7#11: G -> G#

11 -> #11

```
G|12---13---
```

D7#11: Bb -> A

b13 -> 5

```
B|11---10---
```

D7#11: Eb -> D

b9 -> R

```
B|4----3----
```

D7#11: F -> F#

b3 -> 3

```
B|6----7----
```

D7#11: B -> A

13 -> 5

```
B|12---10---
```

D7#11: E -> D

9 -> R

```
B|5----3----
```

G#7alt: A -> G#

b9 -> 1 / tonic 5

```
B|10---9----
```

G#7alt: B -> B#

#9 -> 3

```
B|12---13---
```

G#7alt: D -> C#

b5 -> tonic R

```
B|15---14---
```

G#7alt: E -> C#

#5 -> tonic R

```
B|17---14---
```

G#7alt: F# -> E

b7 -> tonic b3

```
e|14---12---
```

G#7alt: B# -> C#

3 -> tonic R

```
B|13---14---
```

Three-pass improvisation capstone

Pass 1: D7#11 only, resolving to C#m. Pass 2: G#7alt only, resolving to C#m. Pass 3: choose direct or substitute dominant by phrase function; include one full-beat rest every four bars and close on C#m.

Scale Exercises - Section IX, Level 72

Practice progression, error diagnosis, mastery gate, and log

SECTION IX COMPLETION

IDENTIFY THE DOMINANT, THE SHARED TRITONE, AND THE DESTINATION BEFORE PLAYING

Eleven-phase practice progression

- [ ] 1. D Lydian dominant shell and mode
- [ ] 2. Five D Lydian dominant mirrors
- [ ] 3. D7#11 tension cells
- [ ] 4. G#7alt shell and altered mode
- [ ] 5. Five altered mirrors
- [ ] 6. Altered-to-tonic resolutions
- [ ] 7. Shared-tritone shell cells
- [ ] 8. Direct/substitute cadence pairs
- [ ] 9. C#-minor applied phrases
- [ ] 10. Transposition cells
- [ ] 11. Model solo and capstone

Error diagnosis

LOST D7	play D-F#-A-C before adding G#
LOST G#7	play G#-B#-F# before alterations
WRONG C SPELLING	name B# over G#7 and C over D7
SUB DOES NOT RESOLVE	make D fall to C#
ALTERED SOUP	use one alteration and one target
DIRECT/SUB CONFUSION	add the root beneath the shared tritone
FORM LOSS	return to four whole-note chord targets

Section IX mastery gate

- [ ] Play D Lydian dominant and G# altered in all printed routes and name every chord role.
- [ ] Resolve all D7#11 and G#7alt tension cells without losing the destination.
- [ ] Play the shared C-F# tritone in five registers and identify the root by ear.
- [ ] Perform all direct-dominant and tritone-substitute cadence pairs.
- [ ] Play all five C#-minor applied-dominant phrases and all four progression comparisons.
- [ ] Transpose the direct/substitute guide-tone relationship to C, F, A minor, and D minor.
- [ ] Perform the sixteen-bar solo and all three improvisation passes on two separate days.

Practice log

NEXT Section X: chromatic harmony, modulation, odd grouping, repertoire, and full-performance orchestration.

Date	Module / dominant	Destination	D/U	BPM	Clean?	Weak tension / resolution	Next target

Standard tuning. Parent collection: A melodic minor. Dominant functions are named by chord and destination.